

Exploration of Interfaces for Budgeting and Ideation within Group Travelling

2. Description of the experiment

2.1 Introduction and goals

In our experiment we sought out to find the differences, if any, between our mobile application prototype and long-established group travel tools such as a spreadsheet or a group chat. More specifically, we plan to address the following questions:

- Does our budgeting interface reduce the cognitive load that comes with financial distribution compared to spreadsheets?
- Does our idea creation and voting interface reduce the cognitive workload that comes with the ideation process compared to group chat polls?

2.2 Methods

2.2.1 Participants

We studied 8 participants (5 male, 3 female) with an age range of 20 to 24 years old. Participants were recruited through convenience sampling, as we drew from a pool of our university friends. The range of programs studied by the participants encompassed Computer Science, Mechanical Engineering, Pharmacy, Microbiology, and Geology. Each participant went through a pre-screening to ensure that they had some experience with group traveling, and we filtered out those who have not group traveled at least once in the past 2 years. Participants were rewarded with a snack.

Note that in Milestone 3 we stated we would attempt to study 12 participants, but we settled for 8 participants due to time and resource constraints.

2.2.2 Conditions (Interface screenshots found on Appendix)

For the budgeting interfaces:

- Condition A (Baseline): Condition A (Baseline): Participants used a Google Sheets spreadsheet with a popular pre-set budgeting template. The template was edited to only include columns for expense name, cost, the payer, and the financial distribution among the members for that expense. There were also fields tracking the payments each individual member had to make.
- Condition B (Changed): Participants used the budgeting feature in our app. Functionality included expense creation, management of financial distributions for expenses, and viewing and tracking payments made by each member.

For the ideation interfaces:

- Condition A (Baseline): Participants used the polls feature in a Telegram group chat to view and vote on different ideas for activities/destinations. There were conversational chats in between polls in order to mimic a real group chat environment.
- Condition B (Changed): Participants used the ideation feature in our app. Functionality included creating ideas, voting for an idea with a “like or skip” voting system, and viewing all created ideas.

2.2.3 Tasks (Detailed scenarios found on Appendix)

Task 1: Budgeting Task

- Participants were given a scenario where they were a part of a travel group of 5 and the group had to manage final expenses after the trip had ended. There were no pre-loaded expenses in the interface used. Participants had to create 4 expenses, with each expense having different members in the group and/or different financial distributions. Participants then had to list out how much they owed to and were owed by each member. After listing this out, the task ended.

Task 2: Ideation Task

- Participants were given a scenario where they were a part of a travel group of 5 deciding on which activities they were going to do. Participants were shown 10 ideas (with a preset number of votes, as if everybody else in the group had already voted) and had to vote on each idea. Participants were then asked to find the top 2 most liked tasks and the top 2 least liked tasks among the group. After identifying these correctly, the task ended.

After running our pilot study, we made some minor changes to the tasks from Milestone 3. These changes are underlined.

2.2.4 Design

We utilized a within-subjects experimental design for both the budgeting and ideation tasks. We chose this approach to allow each participant to directly compare the two interfaces, as some participants may not have had experience with using both a budgeting app and a budgeting spreadsheet for group travel (similar logic applies to the ideation interfaces as well). Each participant being able to compare the interfaces also offered potential insight during the post-interview that could not be discovered with a between-subjects design. Given our resources and time constraints, a within-subjects design was the most feasible due to the amount of participants we could get.

Due to the within-subjects design, we decided to counterbalance the order of tasks done (budgeting and ideation) as well as which interface was used for each specific task, in order to mitigate order effects. To further reduce order effects, we also made each specific task scenario isomorphic, which makes them slightly different but with the same level of difficulty. For example, when budgeting, the number of group members, names, and overall scenario were the same, but the expenses and financial splits were slightly different. This concept was applied similarly to the ideation task, where the ideas differed but had the same genre and similar location where possible.

2.2.5 Procedure

Upon arrival, participants completed a pre-experiment survey designed to capture their demographics and prior experience with group travel, spreadsheet applications, and group chat polling features.

For each of the two main tasks (budgeting and ideation), participants followed a standardized sequence:

- a. First, we introduced participants to their randomly assigned initial interface and gave them a three-minute unstructured exploration period to familiarize themselves with the environment. This exploration phase helped determine the intuitiveness of the interfaces involved in this experiment.

- b. Following this sandbox phase, participants were given two minutes to understand the scenario for the task. The scenario was given on paper, and participants were encouraged to ask any questions for clarification.
- c. After the scenario phase, participants were asked to start the required task as was listed in **2.2.3**.
 - d. Note that in Milestone 3, we stated we would time the budgeting task, but we elected against this.
- e. Immediately upon task completion, participants filled out a post-task survey to measure perceived mental effort, frustration, and other categories.
 - i. The questionnaire they filled out was the **NASA Task-Load Index (TLX)**, which contains 6 individual categories and used a 20-point scale for measuring these aspects. Using this 20 point scale, the value is multiplied by 5 to get a score from 1-100.
- f. We then offer a 1 minute break to the participants. Afterward, this exact sequence (a two-minute exploration, scenario execution, and immediate post-task questionnaire) was then repeated using the alternate interface.
- g. After completing both scenarios for each category, the researchers conducted a short questionnaire to gather qualitative feedback (refer to Post-Task Questionnaires in the Appendix). Researchers recorded their answers by writing notes based on the participants' responses.

Participants were offered a 5 minute break between the budgeting tasks and the ideation tasks. Once a participant completed both tasks, the researcher conducted a general questionnaire regarding both application tools and baseline tools.

2.2.6 Apparatus

Experiments were held in a small booked room in the university, with one big table and the participants sitting across from us. The spreadsheet portion of the budgeting task was done on a 14" MacBook Pro with a 1920x1080 display resolution with an external mouse connected to it. The rest of the tasks were done on an iPhone 13. The tasks were live recorded through a Discord shared screen so that we could help the participants navigate through the tasks. The NASA TLX assessment was conducted by paper. Researchers used Google Docs to record notes on participant questionnaires. To implement our application interface, we used Figma Make and React to create a mobile-like interface on a website.

Note that in Milestone 3, we stated we would use a digital timer, but we did not need to anymore due to changes in the budgeting task.

2.2.7 Independent and dependent variables

Independent Variables - The interface used (our application, Google Sheets, Telegram)

Dependent Variables - Perceived workload of task (measured with the NASA TLX on a 20-point scale)

During Milestone 3, the time it takes for the budgeting task was included as a dependent variable. For our experiment, it was excluded (refer to Pilot Study).

2.2.8 Hypotheses

Budgeting Task

- **H0:** There is no difference in the perceived cognitive workload that comes with financial distribution when using our interface and a spreadsheet.
- **H1:** Users will report significantly lower cognitive workload when using our interface than those using the spreadsheet.

Ideation Task

- **H0:** There is no difference in the perceived cognitive workload when suggesting and voting on trip ideas between our interface and a group chat poll.
- **H1:** Users will report significantly lower cognitive workload when using our interface to suggest and vote on trip ideas compared to a group chat poll.

Note that in Milestone 3, we had an extra hypothesis related to the time it takes for the budgeting task, but it is now excluded.

2.3 Problems/Limitations

The most obvious limitation is that we were only able to get individuals to play out group scenarios, which is a threat to external validity. The group scenarios we have created are thorough and help visualize a real group planning situation. Playing these scenarios out with individuals can capture individual behaviours in a group that can be applicable to a real group setting, however without truly working in a group with other participants, we cannot fully conclude that the results and behaviours would be the same. For example, a participant may be more engaged when doing our individual task, but when put in a real group setting, they may be more hesitant and less engaged. This is especially present in our ideation phase, where voting and making a decision on an idea would usually require extra discussions with other group members instead of just voting based on a poll or on a screen.

A potential limitation we considered is that participants may not have provided fully accurate answers to our questionnaires due to biases and order effects. For instance, it is possible that due to a combination of our participants being personally close to us and figuring out that they were using an interface that we created, they may have rated our interface more favorably. The participants could have also subconsciously known that we were specifically trying to see if our created application is better than the alternative spreadsheet solution.

It is important to note that counterbalancing may reduce the risk that comes from an order effect, but does not fully eliminate it. For example, in our experiment (particularly the budgeting task), it seems participants may have gotten a better idea of how a given task should be completed after experiencing it with the first interface, which helped them perform better with the second interface. To stay within the scope of our course, we accepted this limitation. For future explorations, we could run a between-subjects experiment in order to reduce this bias.

3. Results

3.1 Pre-Experiment Survey

We conducted a pre-experiment survey in order to find more information regarding participants' group travelling experiences, including spreadsheet usage and group chat polling usage. We had four participants who travelled 1-2 times, one participant in the 3-4 range, and

one in the 6+ range. For spreadsheet proficiency, we had one participant, who was a beginner, six participants who were intermediate, and one advanced. With the group chat polling usage, we had one participant who never used it, one for 'rarely', five for 'occasionally', and one for 'frequently'. The survey structure can be found in the appendix.

3.2 Task Quantitative Results

We analyzed the results using a paired, one-tailed t-test to determine whether our application requires a significantly lower overall workload than the baseline for both tasks. Using our 20-point scale NASA-TLX results, we used the Raw-TLX value (calculated by averaging all the categories in the survey) to reject or accept our null hypothesis. We displayed bar charts to show the individual category results (mental demand, physical demand, etc.) within each task. We are looking for a lower overall score on the application compared to the spreadsheet, which would indicate whether our application has a lower workload when completing the task. Note that for the category of "Performance/Success", a higher score means the participants faced more challenges, while a lower score indicates that the workflow was smoother.

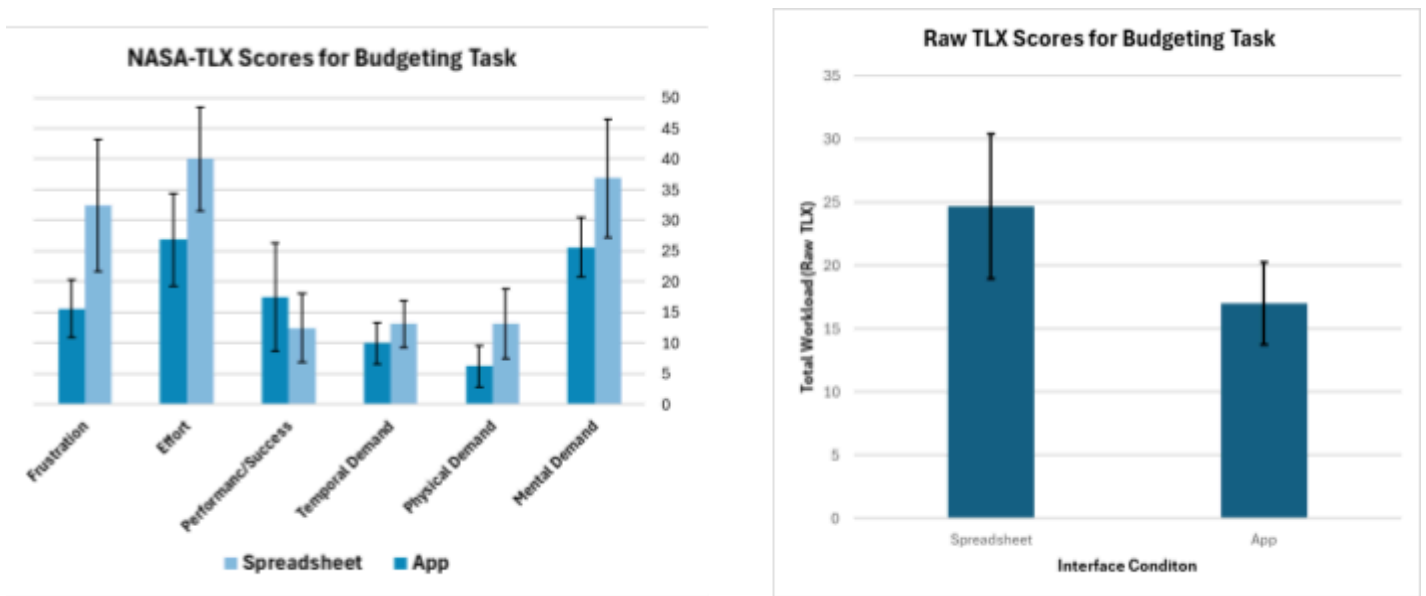


Figure 1: Left: Mean scores of each category in the NASA-TLX for each condition (on a 20-point scale) for the budgeting task

Right: Overall mean score (Raw TLX) for each condition for the budgeting task

*Note: Error bars represent standard error (± 1)

Figure 1 shows that our application resulted in a less overall workload compared to the spreadsheet (Raw TLX mean value: Spreadsheet **24.69 s.e. 5.72** vs. Application **16.98 s.e. 3.25**), however these results did not show to be statistically significant ($t(7) = 1.56$, $p = 0.081$). The figure to compare each category in the survey also displays a total overview of each category within the NASA-TLX. We discovered that our application resulted in less "effort" than the spreadsheet (Effort score: Spreadsheet **40 s.e. 8.45** vs. Application **26.88 s.e. 7.56**, $t(7) = 2.35$, $p = 0.026$). The other categories, however, were found to have no statistical difference. Therefore, we fail to reject our null hypothesis and conclude that we do not have sufficient evidence that the effort required to complete the budgeting task in our application is lower than the effort needed when using a spreadsheet.

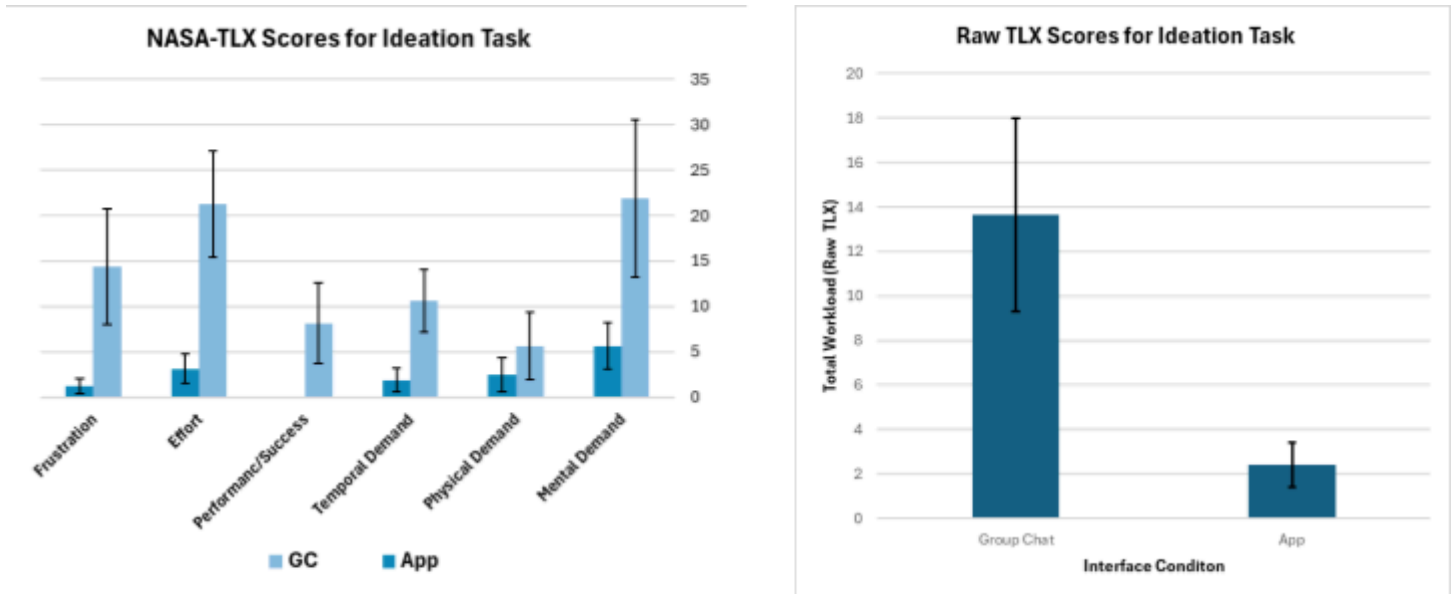


Figure 2: Left: Mean scores of each category in the NASA-TLX for each condition (on a 20-point scale) for the ideation task
 Right: Overall mean score (Raw TLX) for each condition for the ideation task
 *Note: Error bars represent standard error (± 1)

Ideation Task Results

Figure 2 shows that our application resulted in a less overall workload compared to the group chat which is statistically significant (Raw TLX mean value: Group Chat **13.65 s.e. 4.34** vs. Application **2.40 s.e. 1.01**, $t(7) = 3.26$, $p = 0.007$). For 4 out of the 6 categories in the NASA-TLX, we found that our application resulted in less workload that is statistically significant (“Mental Demand”: $p = 0.02$, “Temporal Demand”: $p = 0.008$, “Effort”: $p = 0.002$, “Frustration”: $p = 0.037$). Therefore, we reject our null hypothesis and conclude we have sufficient evidence that participants face lower cognitive workload and frustration when using our application to suggest and vote on trip ideas compared to a group chat poll.

Qualitative Themes and Results

We analyzed participants’ questionnaire responses and observation notes by grouping similar responses into common themes:

Automation Reducing Manual Effort - Participants commonly reported that the application required less manual work, especially for budgeting tasks. Excluding members to split the purchase was frequently mentioned as easier when using checkboxes in the app, while the spreadsheet required manual calculations.

Limited Error Recovery in the Application - Some participants reported that they had difficulty correcting mistakes in application. In particular it was noted that they could not edit or undo entries once submitted.

Group Chat required more navigation and tracking - Participants reported that group chat required more scrolling and manual tracking of information. Polls and decisions were

often spread across multiple messages and as such it sometimes became hard to find what you were actually looking for without things getting lost.

The application supported organization and comparison - For ideation tasks, participants noted that the application made it easier to view and compare ideas. Responses referenced features such as sorting, filtering, and having ideas presented in a single location.

Familiarity with existing tools - Participants also reference prior experiences with spreadsheets and group chats. These tools were described as familiar and commonly used, even when the application was noted as easier for the tasks.

4. Discussion

4.1 Interpretation of our Results

In our results, we found that the application did not result in less cognitive workload than the spreadsheet for the budgeting task. However, for the ideation task, we discovered that the application resulted in less cognitive workload, with four out of the individual categories in the NASA-TLX being statistically significant to conclude our application requiring less workload for those specific categories (“Mental Demand”, “Temporal Demand”, “Effort”, “Frustration”). We also found common patterns, through open coding, within each task and interface combination. We will discuss each task separately to interpret what each result means along with the questionnaire data, as well as flaws within our design that may have impacted our findings.

Budgeting Task

We believe that we did not find workload improvement for our application for a couple of reasons. The biggest reason is that even after our modifications to our budgeting instructions for more clarity, many participants still got confused with interpreting the question, especially when formatting the “final answer” for the task. This occurred multiple times for participants who started with the application and ones who started with the spreadsheet, which introduced noise to our final results. For example, after handing the instructions and giving a brief explanation of the “example answers”, some participants still mistook that example as being part of the whole task. These slight misunderstandings with the interpretations may have resulted in one interface having naturally higher workload scores than the other, which also increases the learning effect going into the next interface. A better exploration phase where we go through an easier, but similar task to inform them how the scenarios will work would reduce these confusions and learning effect, rather than the less controlled exploration phase that was conducted during our study.

Another reason is the statistical analysis that we chose. For budgeting, we did find each “category” of the NASA-TLX favouring our application, with the “Effort” category being statistically significant. Rather than using a t-test which assumes normal distribution and equal variance, it is possible that our results would be statistically significant if we were to use a non-parametric test based on ranks, such as the Wilcoxon Signed-Rank test, which can work better with non-normally distributed data. Since the NASA-TLX is a 20-point scale (equal to a range of 1-100 scores), most participants answered only in the lower half of the scale, but some answered using both halves of the scale. Considering the noise that was mentioned, this can result in the t-test being skewed, but a ranked test being affected less and focused more on one interface being better than the other.

Even after accepting our null hypothesis for this task, we still do believe that our application provides interaction techniques that were preferred compared to our templated spreadsheet

for many different reasons. Through the post-task questionnaires, it is clear that many of the participants liked the application's automation compared to the spreadsheet when distributing the expenses to each member, and when excluding a certain member. This shows that for equal expense distributions, interfaces should support quick "select and unselect" options, like a checkbox, that automatically splits the expense between the members, removing the cognitive load of splitting percentages manually. This does not test situations where there are uneven distributions within a certain expense, but this was a deliberate choice, as a majority of group travellers generally create evenly distributed expenses with other members.

Furthermore, these comments from the participants does not mean that all spreadsheets require more workload overall, as spreadsheet templates for group budgeting are vastly different. For example, our design could have also included a spreadsheet that has the exact same checkbox feature, which could account for the difference that our participants felt when doing the task. Even if the template we picked was fairly popular online, it is difficult to generalize these findings to other popular spreadsheets due to the variety of templates, and the various ways people prefer to use spreadsheets.

This exposes the main strength of our application; it enforces one structured standard that users can follow without having to worry about their own perceived spreadsheet expertise, as well as the aspect of finding and customizing the right template. The app provides a guardrail for calculations that is hard to break compared to a spreadsheet template where formulas and cells can be edited and broken. Even with this guardrail, our application still provides lots of freedom outside of evenly split expenses; participants are able to manually distribute each cost or percentage that a member owes, which our task did not require to complete.

Ideation task

As mentioned, our statistical analysis provides sufficient evidence to reject the null hypothesis. The data demonstrates that our application significantly reduces overall cognitive workload and frustration compared to a standard group chat poll, thus supporting our alternative hypothesis. However, to understand what this difference truly means, we must examine the mechanics of the task itself.

The results were notably lopsided in favor of our application, primarily due to the specific demands of the experimental task. Participants were asked to vote and subsequently identify the most and least popular ideas. In the group chat baseline, this required users to manually scroll through a feed of messages, locate scattered polls, and hold those various poll results in their working memory to compare them. This heavy reliance on working memory and continuous scrolling explains the significant spikes in the "Mental Demand," "Temporal Demand," and "Effort" categories of the NASA-TLX. Our application consolidated this information, removing the cognitive overhead of searching and memorizing.

While the difference is important and validates our design, it is worth acknowledging that the group chat's poor performance was exacerbated by the lack of organizational features. The comparison could be made more robust in future iterations by either improving the task design to reflect more casual, organic voting behavior, or by simulating a more advanced group chat environment (e.g., one equipped with poll filters that aggregate voting results in a single view).

Despite the application proving significantly easier to use, qualitative feedback revealed that lower cognitive workload does not automatically equate to a total user preference. Many users are deeply used to the conversational, free-flowing nature of group chats, which are superior for open discussion and rapid brainstorming. Rather than viewing these interfaces as mutually exclusive, one participant insightfully suggested a hybrid approach. The group chat should remain the space for initial discussion, while our application serves as the authoritative "final

say" for organized voting and idea viewing. This highlights that our tool is more effective as a complementary system of record rather than a complete replacement for group messaging.

Finally, a primary limitation of this study is its execution as a simulated, solo scenario. By testing the interfaces individually, the experiment inherently lacked the complex social dynamics of real group travel planning. While both our prototype and the baseline Telegram interface support anonymous voting to mitigate peer pressure, our application significantly diverges in its presentation of information and interaction design.

Standard group chat polls are often contextually sparse, forcing users to conduct external research and clutter the chat with clarifying messages before making a decision. Conversely, our prototype enforces informed voting by presenting each idea with a mandatory title and description, supplemented by optional images. Furthermore, our application reduces interaction friction by allowing users to utilize a swipe-based gesture interface or explicit like/dislike buttons, accommodating varying user preferences. Because this was a solo evaluation, we could not empirically measure how this combination of rich, embedded context and flexible interaction impacts real-time group consensus. Future studies should test the prototype with synchronized groups to evaluate its effect on social friction and collective decision-making.

4.2 Design Implications

Our results point to a few key design implications. Firstly, systems should reduce manual work. Participants preferred features like checkboxes and automatic calculations instead of entering percentages manually or doing the math themselves, which also helped reduce mistakes. This suggests that common tools, especially spreadsheet templates, should include easier ways to split expenses, such as quick-select or checkbox options alongside manual percentage input.

Secondly, applications need basic editing features. Not being able to fix mistakes was a major issue, even if the app was otherwise easy to use. Features like edit, delete, and undo would help make the system more usable in practice.

Third, tools should combine both structure and communication. The app was better for organizing, while the group chat was better for discussion. A better solution would be to combine the two in one shared space so users don't have to switch between tools. For example, features like voting, filtering, and decision-making could be embedded directly within group chat platforms instead of requiring a separate application.

Finally, many users preferred familiar tools like group chats and spreadsheets. Even when the app was easier, people were still hesitant to switch because they are used to what they already have and their friends use the same tools. This highlights how difficult it is to get users to adopt a completely new application. As a result, new systems should either integrate with existing platforms or stay close to current workflows to make adoption more realistic.

5. Conclusions

Our study investigates conventional methods for managing group travel logistics, such as budgeting and ideation, and introduces a new application prototype to address certain inefficiencies that the conventional methods may lack. Our prototype provided a budgeting tool, focused on validated and fast input methods to try and reduce the cognitive workload that traditional budgeting spreadsheets may have. It also introduced a novel way for groups to approach decision making regarding group ideas and activities; by using a swipe-based voting interface, and organized lists for the ideas, our prototype tries to enhance the

engagement for ideation within a travel group and improve the overall experience. Our study revealed that we did not have sufficient evidence that our application reduced the cognitive workload for budgeting compared to the spreadsheet. Within ideation, however, we found that participants had a significant decrease in cognitive workload compared to using group chats. Many participants still prefer to use the conventional methods instead of the application tools.

We are excited about future explorations within budgeting and ideation methods that our application challenged. For future work, we plan on expanding the scope to simulate a true group scenario to increase the external validity of our interfaces. Future work should also explore how to improve adoption. Even when the application performed better, many participants preferred to continue using the tools they were already using. A possible direction is to design systems that integrate with existing platforms or closely match current workflows (mentioned earlier) to make them easier for groups to adopt.